

Seemab — Detailed Marking Report

Chemistry | exam-03-ch6 — Chapter 6: Stoichiometry | Attempt date: 2026-05-12 | Marked: 2026-05-12
Board of Intermediate and Secondary Education | SSC-I (9th Grade) | Version 1

44.5/46

TOTAL SCORE

96.7%

PERCENTAGE

A+

GRADE

12/12

SECTION A MCQ

24/24

SECTION B

8.5/10

SECTION C (BEST 2)

Section A — Multiple Choice Questions (12 × 1 = 12 Marks) | Score: 12/12 PERFECT

#	Topic	Correct Answer	Seemab's Answer	Result
1	Empirical formula of glucose (C ₆ H ₁₂ O ₆)	B — CH ₂ O	B — CH ₂ O	1/1
2	Molecular mass of naphthalene (C ₁₀ H ₈)	C — 128 amu	C — 128 amu	1/1
3	Formula mass of Mg(OH) ₂	C — 58 amu	C — 58 amu	1/1
4	Particles in one mole (Avogadro's number)	B — 6.022×10 ²³	B — 6.022×10 ²³	1/1
5	Moles of O ₂ molecules in 16 g oxygen	B — 0.5 mol	B — 0.5 mol	1/1
6	Mass of 9.05 moles of ozone (O ₃)	C — 434.4 g	C — 434.4 g	1/1
7	Atoms in 1.25 moles of zinc	B — 7.53×10 ²³	B — 7.53×10 ²³	1/1
8	Formula of aluminium oxide (Al ³⁺ , O ²⁻)	B — Al ₂ O ₃	B — Al ₂ O ₃	1/1
9	Same empirical and molecular formula	C — H ₂ O	C — H ₂ O	1/1
10	Mass of 4 moles of H ₂	A — 8.064 g	A — 8.064 g	1/1
11	Balancing done by changing	C — coefficients	C — coefficients	1/1
12	Ions on both sides not participating	B — spectator ions	B — spectator ions	1/1

9th consecutive perfect MCQ score — MCQ record now 119/120

Section B — Short Questions (11 attempted, best 8 counted) | Score: 24/24

10 questions at 3/3, 1 question at 2/3 (Q2(ix)) — best 8 selected = 24/24

24/24

Q.2(i) — Main Question

Define empirical formula and molecular formula. Give one example of each.

3/3

Full marks

Empirical formula definition precise — "simplest integer ratio" wording matches the textbook. Molecular formula defined correctly as "actual number of atoms." Both examples use H₂O₂ / HO as the answer key specifies. **Clean, structured answer with no padding.**

Q.2(ii) — Main Question

Molecular mass of glucose C₆H₁₂O₆ (C=12, H=1.008, O=16)

3/3

Full marks

6(12) + 12(1.008) + 6(16) = 72 + 12.096 + 96 = 180.096 amu — all steps shown, arithmetic correct, unit stated. **Excellent working shown.**

Q.2(iii) — OR Variant

Differentiate empirical vs molecular formula; two quantities needed to find molecular formula

3/3

Full marks

Table format with definition, other names, "provide information" and example rows — well organised. Both quantities correctly identified (empirical formula mass and molecular mass). "Molecular mass" column correctly labelled. **Table presentation is a consistent strength.**

Q.2(iv) — Main Question

Mass of 0.25 moles of CO₂ (C=12, O=16)

3/3

Full marks

Molar mass CO₂ = 44 g/mol, mass = 0.25 × 44 = 11 g — correct. Formula, substitution, and answer all shown. Minor: final line reads "11 g/mol" instead of "11 g" but context is unambiguous and no mark deducted. **Formula recall is now automatic.**

Q.2(v) — Main Question

Molecules in 0.5 moles of methane (CH₄)

3/3

Full marks

0.5 × 6.022×10²³ = 3.011×10²³ molecules — correct. Avogadro's number recalled correctly. Units and particle type (molecules) stated.

Q.2(vi) — Main Question*Chemical formulas: NaCl, Mg3N2, K2SO4***3/3****Full marks**

(a) NaCl — correct; charges shown as Na^{+1} , Cl^{-1} . (b) Mg_3N_2 — correct; charge balance $3 \times 2 = 2 \times 3 = 6$ explicitly verified. (c) K_2SO_4 — correct; subscripts derived from cross-charge rule. **All three ionic formula derivations shown step-by-step — excellent method.**

Q.2(vii) — OR Variant*Balance $\text{Na(s)} + \text{H}_2\text{O(l)} \rightarrow \text{NaOH(aq)} + \text{H}_2\text{(g)}$ with state symbols***3/3****Full marks**

Atom count on both sides shown. Balanced correctly: $2\text{Na(s)} + 2\text{H}_2\text{O(l)} \rightarrow 2\text{NaOH(aq)} + \text{H}_2\text{(g)}$. State symbols all present and correct. **Atom-count verification table is the right approach.**

Q.2(viii) — OR Variant*Atoms in 0.2 moles of aluminium ($N_A = 6.022 \times 10^{23}$)***3/3****Full marks**

$0.2 \times 6.022 \times 10^{23} = 1.2044 \times 10^{23}$ atoms — exact match to answer key. Formula stated, numbers substituted, answer given with correct scientific notation.

Q.2(ix) — OR Variant [attempted but NOT counted — lowest score, dropped in best-8 selection]*Define ionic equation; write ionic equation for $\text{HCl} + \text{NaOH}$* **2/3****Partial — 1 mark deducted**

Definition correct — "spectator ions removed by net ionic equation" wording acceptable. Molecular equation written. Expanded ionic: $\text{H} + \text{Cl} + \text{Na} + \text{OH} \rightarrow \text{Na} + \text{Cl} + \text{H}_2\text{O}$ — ion charges and (aq) state symbols omitted from all ions throughout (should be $\text{H}^+(\text{aq})$, $\text{Cl}^-(\text{aq})$, etc.). Net equation $\text{H} + \text{OH} \rightarrow \text{H}_2\text{O}$ written without charges. **-1 mark: missing superscript charges and state symbols on all ions in the ionic equation.** This is the only question not at full marks in Section B.

Q.2(x) — OR Variant*Mass of: (a) 1.2 mol K, (b) 0.25 mol steam (H_2O)***3/3****Full marks**

(a) $1.2 \times 39 = 46.8$ g — correct. (b) Molar mass $\text{H}_2\text{O} = 2(1.008) + 16 = 18.016$; $0.25 \times 18.016 = 4.504$ g — correct. **Both parts fully worked.**

Q.2(xi) — OR Variant*Caffeine $\text{C}_8\text{H}_{10}\text{N}_4\text{O}_2$: (a) empirical formula, (b) molecular mass***3/3****Full marks**

Ratio $8:10:4:2 \div 2 = 4:5:2:1 \rightarrow \text{C}_4\text{H}_5\text{N}_2\text{O}$ — correct. Molecular mass: $8(12) + 10(1) + 4(14) + 2(16) = 96 + 10 + 56 + 32 = 194$ amu — correct. **GCD reduction done correctly; a common stumbling point handled well.**

Section C — Long Questions (4 attempted, best 2 counted) | Score: 8.5/10**Q3(i)OR = 4.5/5 | Q3(ii)OR = 4/5 | Q3(iii)OR = 3/5 (dropped) | Q3(iv) = 2.5/5 (dropped)****Best 2: 8.5/10****Q.3(i) — OR Variant [COUNTED]***(a) Define empirical vs molecular formula with examples [2] | (b) Empirical formulas + mol masses: Aspirin, Vinegar, TNT [3]***4.5/5****Near-perfect — 0.5 mark deducted**

(a) 2/2: Empirical formula and molecular formula both defined correctly. Benzene ($\text{C}_6\text{H}_6 \rightarrow \text{CH}$, ratio 1:1) and $\text{H}_2\text{O}_2 (\rightarrow \text{HO})$ both used correctly as examples. Table structure clean.

(b) Aspirin ($\text{C}_9\text{H}_8\text{O}_4$): Seemab wrote " $\text{C}_4\text{H}_2\text{O}_4$ " as the empirical formula. Correct empirical formula is $\text{C}_9\text{H}_8\text{O}_4$ (GCD of 9, 8, 4 = 1, so it is already in simplest form). **-0.5 mark: incorrect empirical formula for aspirin — likely confused the 9:8:4 ratio with a halving that does not exist.**

Molecular mass of aspirin: $9(12) + 8(1.008) + 4(16) = 108 + 8.064 + 64 = 180.064$ amu — essentially correct (answer key uses $\text{H}=1$ giving 180 amu; use of $\text{H}=1.008$ is acceptable). Full credit.

Vinegar ($\text{CH}_3\text{COOH} = \text{C}_2\text{H}_4\text{O}_2$): Empirical CH_2O — correct. Mol mass 60.024 amu — correct.

TNT ($\text{C}_7\text{H}_5\text{N}_3\text{O}_6$): Empirical $\text{C}_7\text{H}_5\text{N}_3\text{O}_6$ (GCD=1) — correct. Mol mass 227.04 amu — correct.

Q.3(ii) — OR Variant [COUNTED]*(a) Moles in: 2.4g He, 250mg C, 40g S [3] | (b) Mass of: 75 mol H_2 , 0.15 mol H_2SO_4 [2]***4/5****Strong — 1 mark deducted**

(a) He: $2.4 \div 4 = 0.6$ mol — correct. mg→g conversion explicitly shown ($250 \div 1000 = 0.25$ g). **Good habit to show unit conversion.**

(a) Carbon: Seemab used molar mass 14 for carbon. Carbon's molar mass = 12 g/mol, not 14. Mass 14 is nitrogen's molar mass. Seemab calculated $0.25 \div 14 = 0.0178$ mol; correct answer is $0.25 \div 12 = 0.0208$ mol. **-1 mark: incorrect molar mass for carbon (14 used instead of 12) — elemental molar masses must be memorised precisely.**

(a) Sulphur: $40 \div 32 = 1.25$ mol — correct.

(b) H_2 : Molar mass = $2(1.008) = 2.016$; $75 \times 2.016 = 151.2$ g — correct.

(b) H_2SO_4 : Molar mass $2(1.008) + 32 + 4(16) = 98.016$; $0.15 \times 98.016 = 14.7024$ g — correct and fully worked.

Q.3(iii) — OR Variant [NOT counted — 3rd best]
(a) Define ionic equation + spectator ions [2] | (b) Ionic equations for 3 reactions [3]

3/5

Not counted — 2 marks lost in (b)

(a) 2/2: Both ionic equation and spectator ion definitions correct and complete.

(b) Mg + H2SO4: Expanded ionic correctly identifies SO4²⁻ as spectator. However the net ionic equation written was "Mg + H2 → MgH2" — this is incorrect. MgH2 is magnesium hydride, not a product. Correct net ionic: Mg(s) + 2H⁺(aq) → Mg²⁺(aq) + H2(g). **-1 mark: wrong net ionic product — Mg²⁺ and H2(g) are the products, not MgH2.**

(b) AgNO3 + NaCl: Ag + Cl → AgCl — correct. Na⁺ and NO3⁻ correctly identified as spectators. Full credit.

(b) Zn + 2HCl: Written as "Zn# + H2 → ZnH2" — same error as reaction 1. Correct net ionic: Zn(s) + 2H⁺(aq) → Zn²⁺(aq) + H2(g). **-1 mark: Zn²⁺ and H2(g) are products — ZnH2 does not form in this reaction.**

Pattern note: Seemab consistently confuses "Zn + H2" as forming ZnH2 rather than yielding Zn²⁺ + H2(g). The hydrogen gas is released, not bonded. This is the same ionic equation misunderstanding seen in Q2(ix) (missing charges).

Q.3(iv) — Main Question [NOT counted — lowest score]
0.5g K + H2O: (i) Balance [1] (ii) Moles of K [2] (iii) K atoms [2]

2.5/5

Not counted — significant errors in (ii) and (iii)

(i) Balancing: 1/1. 2K + 2H2O → 2KOH + H2 — correct and fully checked.

(ii) Moles of K: 1.5/2. Molar mass K = 39 g/mol ✓. moles = 0.5/39 = 0.0128 mol ✓. Value correct. **-0.5 mark: final answer written as "0.0128 mol/g" — the unit is mol, not mol/g. Incorrect unit on the final answer.**

(iii) K atoms: 0/2. Seemab wrote: "39 × 6.022×10²³ = 234.858×10²³". This multiplies the molar mass (39) by NA instead of multiplying the moles (0.0128) by NA. **-2 marks: formula applied incorrectly — atoms = moles × NA = 0.0128 × 6.022×10²³ = 7.71×10²¹ atoms. Using the molar mass instead of the mole count is a fundamental conceptual error.**

STRENGTHS THIS ATTEMPT

- MCQ: 9th consecutive perfect score (12/12) — outstanding recall
- All mole/mass calculations in Section B fully correct and shown
- Ionic formula derivation (NaCl, Mg3N2, K2SO4) — step-by-step method clear
- mg to g unit conversion shown explicitly (Q3(ii)a He)
- Molecular mass with H=1.008 precision maintained throughout
- Caffeine empirical formula GCD reduction done correctly
- Equation balancing with atom-count verification — solid method

WEAKNESSES TO REVISE

- Ionic equations — ion charges and state symbols missing:** H⁺(aq), Cl⁻(aq) etc. must be written with charges and (aq) — seen in Q2(ix) and Q3(iii)
- Net ionic products:** "Mg+H2→MgH2" error — hydrogen is released as H2(g), not bonded to metal as a hydride in aqueous reactions (Q3(iii))
- Moles × NA not molar mass × NA:** Q3(iv)(iii) used M=39 instead of n=0.0128 — the atom-count formula must use moles, not molar mass
- Carbon molar mass = 12, not 14:** Q3(ii)a — Nitrogen is 14, Carbon is 12 — a memorisation gap
- Aspirin empirical formula:** C9H8O4 is already in simplest ratio (GCD=1); do not halve when no common factor exists

TOP 3 REVISION TARGETS BEFORE NEXT CHEMISTRY EXAM

- 1. Ionic equation notation:** Practice writing ALL ions with correct superscript charges (H⁺, OH⁻, Mg²⁺, SO4²⁻) and state symbols (aq). Write 5 full ionic equations from scratch.
- 2. atoms = moles × NA (not M × NA):** The three-step chain is: mass → moles (÷M) → atoms (×NA). Drill this chain with 5 examples until it is reflexive.
- 3. Elemental molar masses:** Flash-card drill — C=12, N=14, O=16, S=32, K=39, He=4. C and N are one apart and frequently confused.

Chemistry — Prior Attempts Comparison

#	Date	Exam	Score	%	MCQ	Sec B	Sec C	Grade	Notes
1	12 Apr 2026	exam-01-ch1-2	64/65	98.5%	12/12	33/33	19/20	A+	Near-perfect; 1 mark off Sec C
2	17 Apr 2026	exam-02-ch3-5	59.5/65	91.5%	12/12	29.5/33	18/20	A+	Periodic trends direction; bonding gaps
3	12 May 2026	exam-03-ch6 NEW	44.5/46	96.7%	12/12	24/24	8.5/10	A+	Ionic eq. notation; C molar mass slip; atom-count formula error

Note: exam-03 is 46 marks total (not 65); percentage comparison is like-for-like. Strong bounce-back after exam-02 dip.